

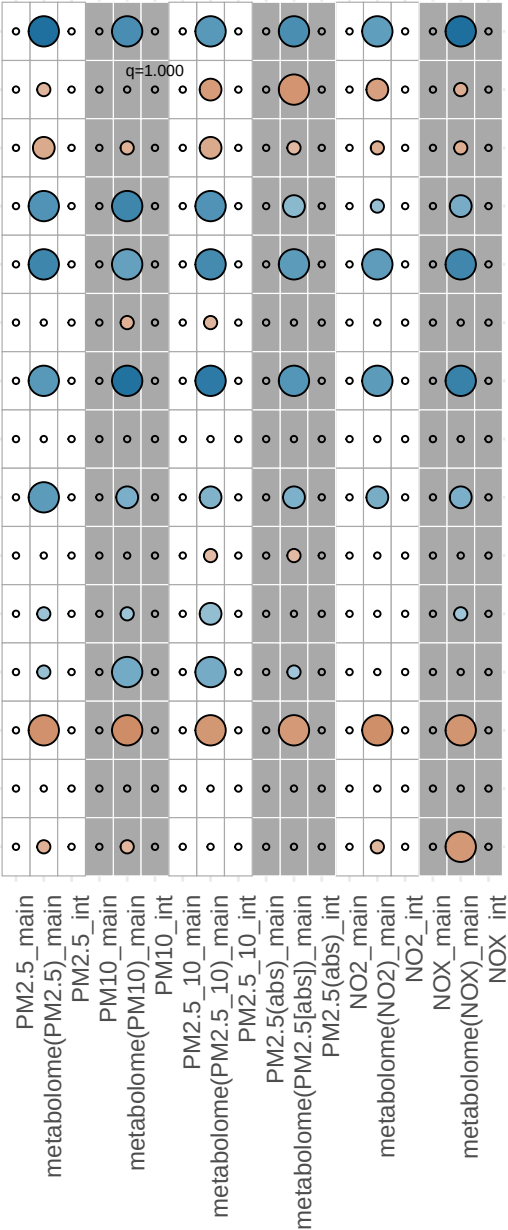
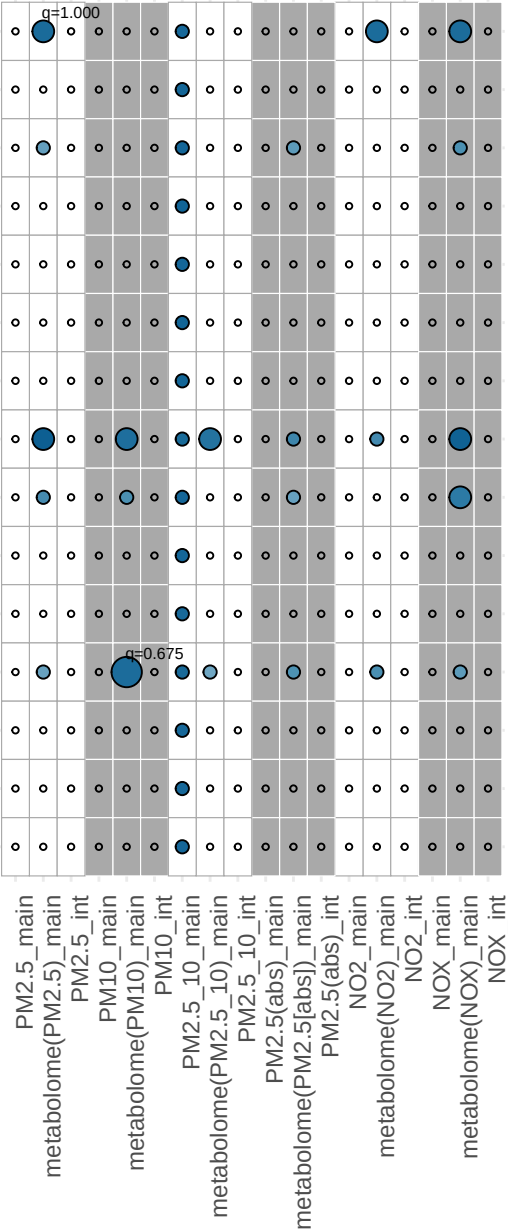
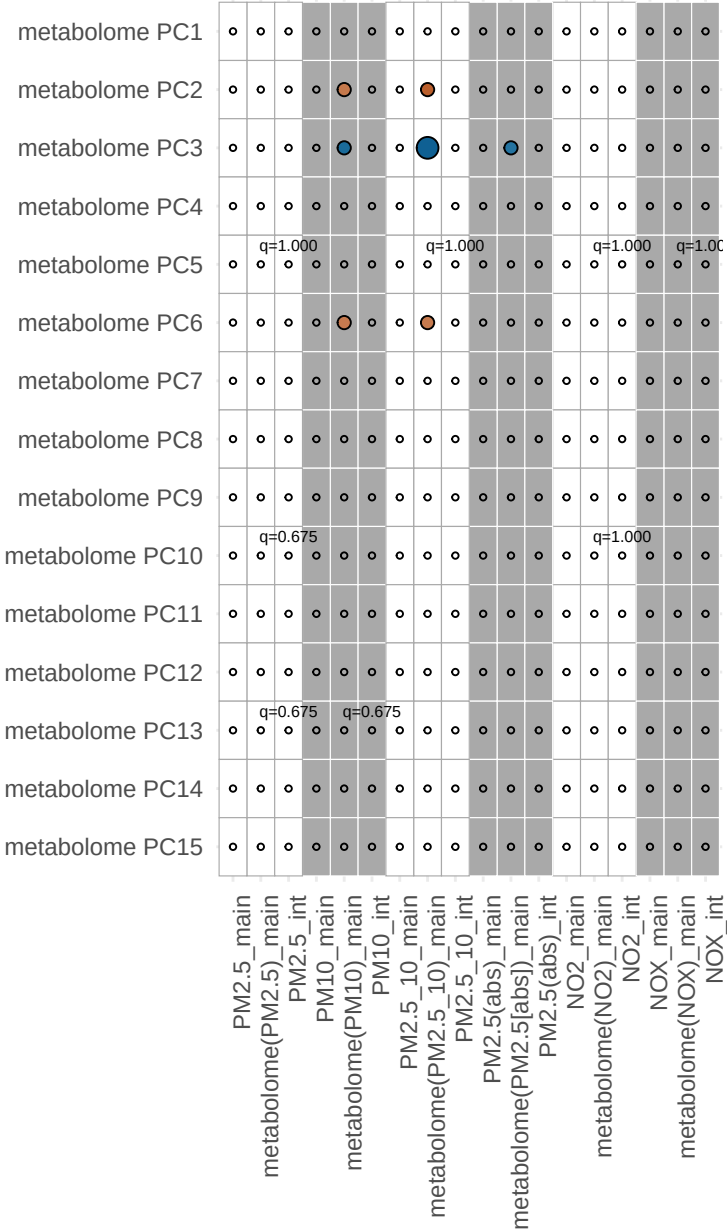
Heatmap (Weibull regression models – Air pollution markers and plasma metabolome results)

Healthy to Parkinson's Disease

Healthy to Dementia

Healthy to Mortality

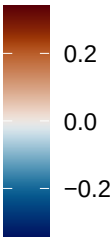
Metabolome Principal Components



Significance
main effects: nominal p
interactions: BH q

- $\bullet$ < 0.001
- $\bullet$ < 0.01
- $\bullet$ < 0.05
- $\bullet$ < 0.10
- $\circ$ $= 0.10$

β coefficients



Air pollution markers

Exposures are binary above/below median; metabolome PCs are z-scored. Interaction-cell significance uses Benjamini-Hochberg q-values with $q < 0.10$, after a nominal interaction screen of $p < 0.10$. If the Stata export only contains stars, q-values